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Binary Alloys Hot Paper

Metastable Liquid Properties and Surface Flow Patterns of Ultrahigh Temperature Alloys Explored in Outer Space

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Abstract: The metastable liquid properties and chemical bonds beyond 2000 K remain a huge challenge for ground-based research on liquid materials chemistry. We show the strong undercooling capability, metastable liquid properties and surface wave patterns of refractory Nb–Si and Zr–V binary alloys explored in space environment. The floating droplet of $Nb_{82.7}Si_{17.3}$ eutectic alloy superheated up to 2338 K exhibited an extreme undercooling of 437 K, approaching the $0.2T_E$ threshold for homogeneous nucleation of liquid-solid reaction. The microgravity state endowed alloy droplets with nearly perfect sphericity and thus ensured the high accuracy to determine metastable undercooled liquid properties. A special kind of swirling flow was induced for liquid alloy owing to Marangoni convection, which resulted in the spiral microstructures on $Zr_{64}V_{36}$ alloy surface during liquid-solid phase transition. The coupled impacts of surface nucleation and surface flow brought in a novel olivary shape for these binary alloys. Furthermore, the chemical bonds and atomic structures of high temperature liquids were revealed to understand the liquid properties in outer space circumstances.

Introduction

Liquids are so popular in the natural world and our daily life, such as water, alcohol, syrup, acetic acid, and molten steel, etc.[1–8] However, there are still many mysteries for liquids at extraordinary environments, like superhigh or ultralow temperatures, and hyporgravity or microgravity.[9–15] Especially, some novel properties and flow behaviors for a liquid in outer space has aroused great research interests in the fields of materials and chemistry.[16–22] Not only the liquid state is a fundamental problem in physical chemistry, surface chemistry and high temperature chemistry,[23–27] but also the research of liquid properties and the chemical bonds in outer space will play a significant role in multidisciplinary research connecting chemistry with space physics and metallic materials.[28–31]

The metastable liquid thermophysical properties and surface flow patterns of refractory alloys with liquidus temperatures (T_L) beyond 2000 K face the challenges of high-temperature chemistry reactivity, intense volatilization, and droplet deformation. Although ground-based aerodynamic /electromagnetic /acoustic /electrostatic levitation technologies[32–40] have been developed to resolve these problems, the inevitable gravity field introduces natural convection and droplet deformation, which limit the properties determination of metastable and ultrahigh temperature liquid alloys. Previously, an electromagnetic levitation device was installed in the International Space Station (ISS), which paves a new avenue for the long-term stable microgravity and containerless environment.[41–44] An electrostatic levitation (ESL) furnace, developed by Japan Aerospace Exploration Agency, was launched to ISS, which mainly intended to handle the molten oxide materials that are difficult to be processed on the ground.[45–47] The space-based ESL in China Space Station (CSS) completed its construction and has been in operation since 2021.[48] This brings a golden chance to perform space experiments on ultrahigh temperature liquid alloys. Before this, our group had conducted ground-based researches for more than thirty years. Accordingly, both of the space-based and ground-based experiments could be conducted for the same alloy system, and detailed researches became available.

Refractory Nb–Si and Zr–V alloys, known for their high-temperature resistance and lightweight properties, hold great potential applications for aerospace engines, space exploration, advanced superconductors, and nuclear power plants. Studying their thermophysical properties and unique surface microstructures in a microgravity environment is crucial for advancing rapid solidification theory and developing high-performance alloys, especially for eutectic compositions. Here, we select refractory $Nb_{82.7}Si_{17.3}$ alloy and Zr–V near-eutectic alloys to conduct metastable liquid properties and surface flow patterns in a superwide temperature range of 1400–2400 K in outer space. Especially, we try our best to explore the extreme undercooling capability of the floating droplets, and then, to experimentally determine the fundamental properties, such as density and cooling rate. Meanwhile, some extraordinary droplet behaviors are paid attention under the effect of surface flow patterns and liquid/solid phase transition in microgravity. Furthermore, the chemical bond and atomic structures of high temperature liquids were revealed by Molecular

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Dynamics (MD) based on Deep Neural Networks (DNN) to understand the liquid properties in outer space, where we overcame the great difficulties of atomic potential scarcity for MD and large atom system for the first principle.

Results and Discussion

Firstly, we measure that the CSS in orbit provides a gravity at $2.603\times10^{-5}\ g_0$, where g_0 is the gravitational acceleration (g) on the ground (Figure 1a). This indicates that there is a wonderful and stable microgravity environment aboard CSS. The space-based ESL employs four laser beams, which are distributed around the sample at the vertices of a regular tetrahedron (Figure S1). The main difference between ESL parameters on CSS and the ground is the levitation voltage. On the ground, a levitation voltage of over 10^4 V/cm is required, while aboard the CSS, the ESL only needs a voltage of less than 1.0×10^3 V/cm. We employed the thermal processing with the aid of the thermal control strategy aboard CSS, and the $Nb_{82.7}Si_{17.3}$ alloy was heated to an ultrahigh temperature 2338 K in CSS, which is 149 K higher than the eutectic temperature T_E of 2189 K (Figure 1b and Figure S2a). Then, the alloy was cooled to a metastable state with a temperature of 1752 K, reaching the largest undercooling of 437 K (20 % of T_E). We find that this value approaches the threshold for the homogeneous nucleation, indicating a strong undercooling capability under microgravity.[49–50] This will greatly help to explore liquid properties and its phase transition under extremely undercooled state. Moreover, under an extremely undercooled state, there is a significant thermodynamic driving force for the alloy melt to initiate nonequilibrium liquid-solid phase transformation. This leads the nucleation and growth rates of crystals to surpass their traditional limits, thereby inducing these alloys to exhibit unusual morphologies and microstructural solidification patterns. In order to ensure a high accuracy of liquid properties, the projected images of the $Nb_{82.7}Si_{17.3}$ alloy are employed to evaluate its sphericity. The projection of $Nb_{82.7}Si_{17.3}$ prealloy (prepared by ESL on the ground, more details see Supporting Information) approximated to an oval. It appears as a circle after the alloy was melted. The standard circles have the same surfaces as the alloy projections are superimposed on the projections (Figure 1c). The circular deviation (Δr, Supporting Information, Equation 1) of the floating alloy droplet surface is calculated over different directions by projection radius $r(\theta)$ and standard circle radius r_0. The largest Δr of prealloy reaches 6.72×10^{-2} mm, which is 4.83 % of the r_0. However, the projection of liquid $Nb_{82.7}Si_{17.3}$ alloy is close to a circle (Figure S2b). It is notable that the maximum Δr in space is only 2.6×10^{-2} mm for the same droplet size, which is much less than 0.27 mm on the ground.[51] The temperature distributions and flow patterns of the alloy droplet under

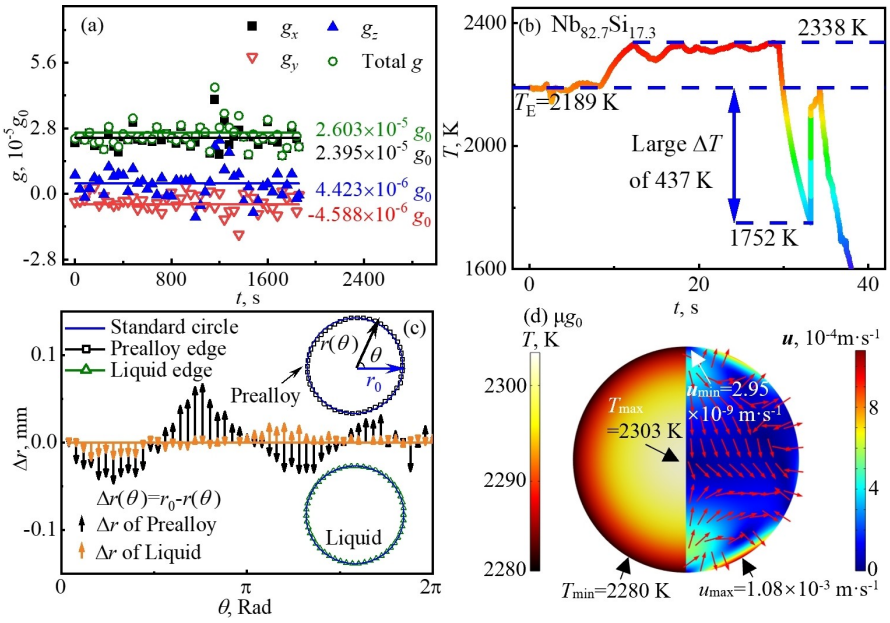

Figure 1. Undercooling capability of liquid $Nb_{82.7}Si_{17.3}$ alloy aboard China Space Station (CSS). (a) The average microgravity level (monitored by a gravity measurement instrument installed on the space station) in the x-direction is 2.395×10^{-5} g_0, in the y-direction is -4.588×10^{-6} g_0, in the z-direction is 4.423×10^{-6} g_0, and the overall average microgravity level is 2.603×10^{-5} g_0 throughout the entire experimental process. (b) Temperature-time curve (measured by an infrared thermometer) with the maximum undercooling 437 K under microgravity. Heating and maintaining an extremely high temperature liquid state (T_{max} = 2338 K) for 15 seconds, followed by natural cooling to achieve a large undercooling of 437 K (20% of T_E). (c) The circular deviation of floating prealloy surface is notably pronounced, but after undergoing heating and melting, the circular deviation of the floating droplet surface becomes comparatively minimal. (d) The temperature distributions and flow patterns under microgravity are calculated by using Finite Element Method. In the cooling process, the temperature field inside the droplet exhibits a gradually decreasing radial distribution from the center to the surface (the central T_{max} = 2303 K, the surface T_{min} = 2280 K), while the flow field is very weak (u_{max} = 1.08×10^{-3} m s^{-1}, u_{min} = 2.95×10^{-9} m s^{-1}).

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microgravity condition were calculated by using the Finite Element Method (Supporting Information, Equation 7–12). In the cooling process, the temperature decreases gradually with the isotherm along the radial direction from the droplet center to the surface (Figure 1d). In addition, there is tiny flow inside the droplet and its flow velocity u is less than $1.08\times10^{-3}\,m\,s^{-1}$. However, the flow field of levitated droplet on the ground is almost two orders larger than that in the microgravity environment.[52] Under the microgravity condition, the disturbances caused by gravity and electrostatic field significantly diminish, resulting in a weaker internal flow within droplet. It is therefore possible to increase the undercooling values of the droplet.

We conclusively think that the Nb–Si alloy has a considerable undercooling capability in outer space, which will cause a significant crystal growth velocity, leading to a particular solidified shape. On the one hand, the decreasing mass transport velocity and solidification time produce a strongly curved solid phase during the nucleation process. A strong mass transportation would occur near the nucleation sites inside the droplet, due to the density difference between liquid and solid $Nb_{82.7}Si_{17.3}$ alloys(Figure 2a and 2e). The large mass transportation velocity and multiple nucleation sites distribute on the surface of liquid alloys at small undercoolings produce a 3-dimensional shrinkage in the solidification process. Thus, the solidified alloys exhibit as spheres for small undercoolings. However, for large undercooling conditions, the liquid viscosity at a large undercooling is nearly increased by 300 % compared with viscosity at T_E,[52–53] leading to a greatly decreasing of the mass transportation. Meanwhile, based on the experimental data on the ground, at a large undercooling of 437 K, the migration velocity of the liquid-solid interface is beyond $100\,mm\,s^{-1}$, approximately 40 times greater than the $2.5\,mm\,s^{-1}$ observed at a small undercooling of 87 K. The mass transportation could not meet the growth of solid phase, which results in the initially generated solid phase exhibiting a large curvature compared with original droplet edge in nucleation process. On the other hand, due to the effect of interfacial tension, the liquid at the liquid-solid interface tends to maintain a small curvature during crystal growth. Therefore, the shape of the residual liquid alloys would not be spherical. Afterwards, the rapid migration of liquid-solid interface prevents the complete relaxation of residual liquid. As a result, the droplet curvature in growth process is smaller than that of the original droplet edge. Interestingly, a combination of these two factors forms the olivary macro-morphology at large undercoolings under microgravity. According to the aforementioned analysis, we can successfully obtain solidified $Nb_{82.7}Si_{17.3}$ alloys with obvious olivary shape due to the significant undercooling capability of the Nb–Si alloy. In the space experiments, regardless of the magnitude of undercooling, there is a distinct growth orientation starting from the nucleation site toward the equatorial direction or the center of the droplet (Figures 2c and 2g), for the spherically symmetric temperature and concentration fields. Noticeably, the solidified alloys with an undercooling of 437 K formed an olivary macro-morphology along the growth direction as the major axis (Figure 2f), with the synergistic effects of microgravity and containerless conditions. As we can see, there are many smooth eutectic cells on the surface with an 87 K undercooling in Figure 2b, which are completely consisted of (Nb) and Nb_3Si eutectic microstructures (Figure 2d). However,

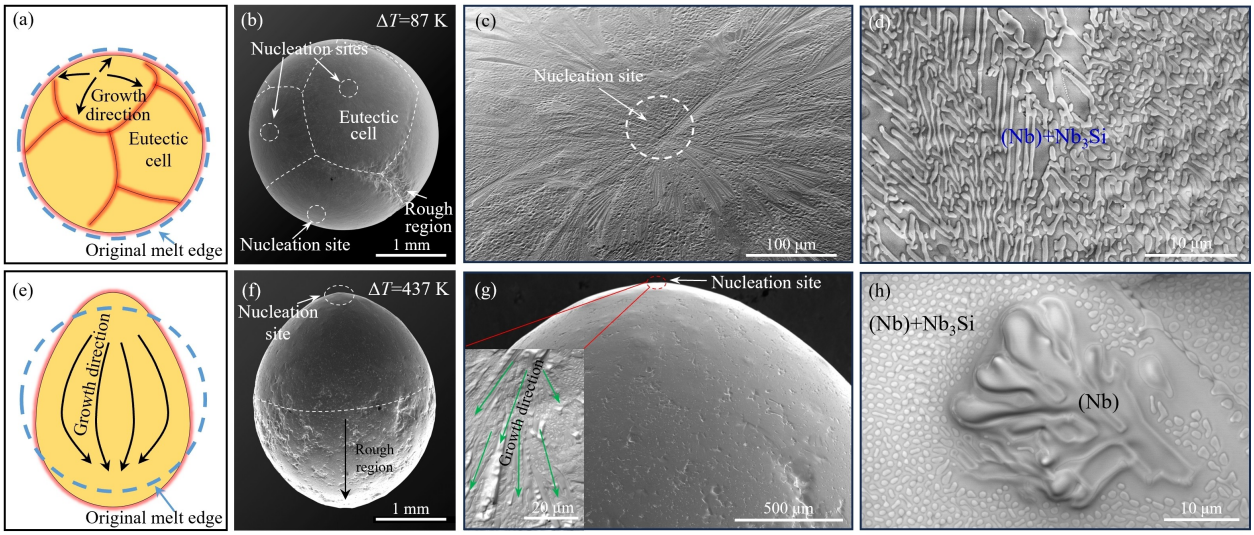

Figure 2. Olivary shape of $Nb_{82.7}Si_{17.3}$ alloy samples solidified aboard CSS. (a) Sketch of a spherical sample surface for small undercoolings because multiple nucleation sites are distributed on the surface in the solidification process. (b) The spherical alloy sample (the average diameter is about 2.736 mm) exhibits multiple smooth eutectic cells on the surface with an 87 K undercooling. (c) Microstructure of a typical nucleation site in (b). (d) The microstructure at eutectic cells on the surface with 87 K undercooling. (e) Sketch of an olivary shaped sample surface for large undercoolings due to fewer nucleation sites, higher liquid viscosity, and higher liquid-solid migration velocity during rapid solidification. (f) Olivary shaped sample (olivary shape with a major axis of 2.939 mm and a minor axis of 2.621 mm) obtained at 437 K undercooling. (g) The microstructure of the nucleation site in (f). (h) (Nb) primary dendrite and eutectic form on the surface at 437 K undercooling.

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not only typical eutectic microstructures but also primary phases are formed (Figure 2h) in the rough region solidified with 437 K undercooling (Figure 2f). Moreover, by leveraging the microgravity conditions in the space station, we can achieve a high undercooling of larger-sized alloy droplets. Through the control of undercooling, we can obtain olivary shaped solidified alloys with different curvatures in outer space, which could be used to directly produce various olivary shaped samples.

Based on the above, we know that alloy droplets exhibit lower deformation rate and higher sphericity, due to the significant disappearance of gravity in the space station. It would achieve higher accuracy, because of the principle of density measurement on space-based ESL.[54–55] After the samples were levitated in the predetermined position, a density camera with UV backlights was employed to capture their clear projections at the frequency of 100 Hz. The sample edge is fitted by the sixth-order Legendre polynomial, and the volume V of the droplet could be obtained by the integration of the Legendre polynomial[56] (Supporting Information, Equation 2–4). Then, the density ρ can be determined. Here, we measured the thermophysical properties of liquid Nb–Si and Zr–V alloys aboard CSS and compared the results obtained on the ground (labeled as CSS and GND). The density ρ of liquid $Nb_{82.7}Si_{17.3}$ alloy was obtained by space-based and ground-based ESL (Figure 3a). It can be seen that the density decreases linearly with temperature. The density at T_E is determined to be 7.42 g cm^{-3} in CSS. Its temperature coefficient is derived as -2.19×10^{-4} g cm^{-3} K^{-1}. The density measured on the ground at T_E is 7.32 g cm^{-3}, which is 1.35 % smaller than that in space. The temperature coefficient of density obtained on the ground is larger. The liquid densities of eutectic $Zr_{57}V_{43}$, hypoeutectic $Zr_{64}V_{36}$, and hypereutectic $Zr_{50}V_{50}$ alloys were also determined aboard CSS (Figure 3b–d). The $Zr_{57}V_{43}$ eutectic alloy experienced four thermal cycles. The density measured in the four thermal cycles are in good agreements, indicating perfect stability of density measurement. The density of liquid $Zr_{57}V_{43}$ alloy in CSS is 6.12 g cm^{-3} at T_E, smaller than the value of 6.19 g cm^{-3} measured by ground-based ESL.[56] The temperature coefficient of density in CSS is lower than the values on the ground. The density of liquid $Zr_{64}V_{36}$ alloy at T_L is 6.13 g cm^{-3}, smaller than the value of 6.16 g cm^{-3} measured on the ground.[56] The temperature coefficient of the liquid density in CSS agrees well with the ground value. The densities of liquid $Zr_{50}V_{50}$ alloy at T_L in space and ground environment are 6.08 g cm^{-3} and 6.15 g cm^{-3}.[56] Their temperature coefficients have a 6.98 % deviation. The density deviation at T_L is less than 1.35 % for the three Zr–V alloy. And the liquid densities of $Zr_{64}V_{36}$, $Zr_{57}V_{43}$ and $Zr_{50}V_{50}$ alloys decreases as the content of V increases (Figure S3). Moreover, the density measurement errors on the ground-based ESL are estimated to be 1.0 % for Nb–Si alloys and 0.9 % for Zr–V alloys, while at CSS they are 0.6 % for Nb–Si alloys and 0.5 % for Zr–V alloys.

The alloy droplets underwent pure radiative cooling in the cooling process aboard CSS. The liquid cooling rate (R_c) and the ratio of specific heat to emissivity (ξ, Supporting Information, Equation 5 and 6) of $Nb_{82.7}Si_{17.3}$ and $Zr_{57}V_{43}$ alloys are measured aboard CSS. The ξ of $Nb_{82.7}Si_{17.3}$ alloy at T_E obtained is 143.56 J mol^{-1} K^{-1} (Figure 4a). The R_c of liquid $Nb_{82.7}Si_{17.3}$ alloy decreases from 239 to 97 K s^{-1} with

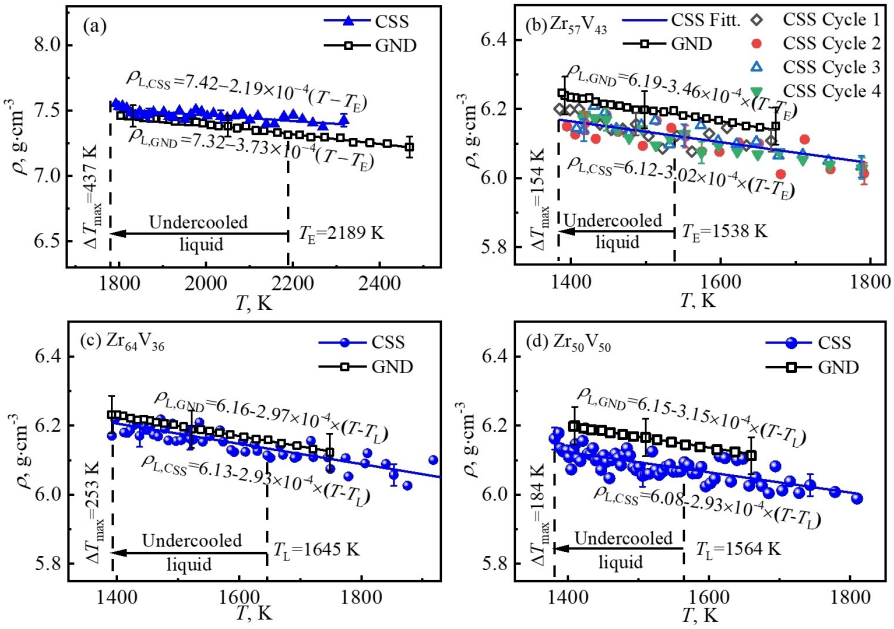

Figure 3. Metastable liquid densities of refractory Nb–Si and Zr–V alloys measured on the ground and aboard CSS. (a) The densities of liquid $Nb_{82.7}Si_{17.3}$ alloy droplet exhibit a linear relationship with temperature. The density at T_E is determined to be 7.42 g cm^{-3} in CSS which is 1.35 % larger than that on the ground, and its temperature coefficient is derived as -2.19×10^{-4} g cm^{-3} K^{-1}. The liquid densities of eutectic $Zr_{57}V_{43}$ (b), hypoeutectic $Zr_{64}V_{36}$ (c), and hypereutectic $Zr_{50}V_{50}$ (d) alloys have linear relationships with temperature for one or more thermal cycles which are in good agreements, indicating perfect stability of density measurement under microgravity.

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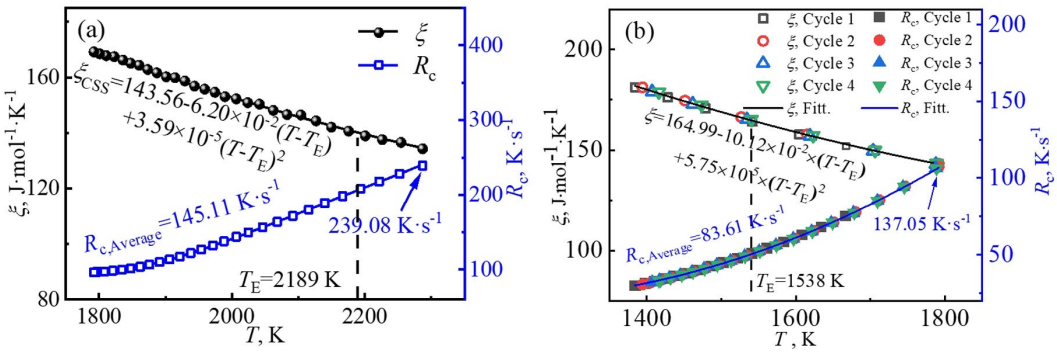

Figure 4. The ratio of specific heat to emissivity and liquid cooling rate measured aboard CSS. (a) The ratio of specific heat to emissivity and temperature of $Nb_{82.7}Si_{17.3}$ alloy exhibits a quadratic relationship as $\xi_{CSS}=143.56–6.20\times10^{-2}(T\text{-}T_E)+3.59\times10^{-5}(T\text{-}T_E)^2$, while the liquid cooling rate and temperature exhibit a positive correlation ($R_{c,max}=239.08\ K\,s^{-1}$ and $R_{c,Average}=145.11\ K\,s^{-1}$). (b) The ratio of specific heat to emissivity and temperature of $Zr_{57}V_{43}$ alloy exhibits a quadratic relationship as $\xi_{CSS}=164.99–10.12\times10^{-2}(T\text{-}T_E)+5.75\times10^{-5}(T\text{-}T_E)^2$, while the liquid cooling rate and temperature exhibit a positive correlation ($R_{c,max}=137.05\ K\,s^{-1}$ and $R_{c,Average}=83.61\ K\,s^{-1}$) for 4 thermal cycles.

decreasing temperature and has an average value of 145 $K\,s^{-1}$. It shows a quadratic relationship with temperature. The ξ and R_c of liquid $Zr_{57}V_{43}$ alloy during the four thermal cycles are shown in Figure 4b, respectively. The ξ of $Zr_{57}V_{43}$ alloy at T_E obtained is 164.95 $J\,mol^{-1}K^{-1}$. The R_c of liquid $Zr_{57}V_{43}$ alloy decreases from 137 $K\,s^{-1}$ to 30 $K\,s^{-1}$ with decreasing temperature. The ξ are fitted by a quadratic Equation of temperature as:

$$\xi = \xi_L + A(T - T_L) + B(T - T_L)^2,$$

where ξ_L represents the ratio at T_L and A, B are constants (Figure 4). From above, the fundamental liquid properties could be applied to guide the researches on liquid flow, droplet oscillation, heat transfer, and surface microstructure.

On the basis of metastable liquid properties, the surface flow patterns role on the solidified surface microstructure could be explored. We know that the microgravity has an effect on the droplet dynamics in outer space, which is quite different from the ground conditions. Thermal induced Marangoni effect exists on the surface of the floating droplet. We have computed the flow field of a $Zr_{64}V_{36}$ alloy droplet under microgravity (Figure 5a). The disappearance of natural convection results in a reduction in the flow velocity within the droplet ($u \leq 1.06\times10^{-3}\ m\,s^{-1}$). And, the Marangoni effect leads to the existence of rich internal flow within the droplet. Furthermore, if there exists a point disturbance (the nucleation point), the solid-liquid interface may experience relative vibrations due to the disparities in their properties (Figure 5b). While these vibrations may be ignored in high-velocity flow fields, they can propagate in the form of waves in weak flow fields. Additionally, if the crystal grows along the surface and the growth rate aligns with the vibration frequency, this should be preserved on the surface.[57] We found visible surface ripples with a wavelength of 25.8 μm on the surface of the solidifed $Zr_{64}V_{36}$ alloy in CSS (Figure 5c). Surface ripples originate from a vertex at surface as concentric circles, and fade at sample equator. Not only the macro shape with millimetre scale exhbits concentric ripples, but also spiral microstructure[58] is

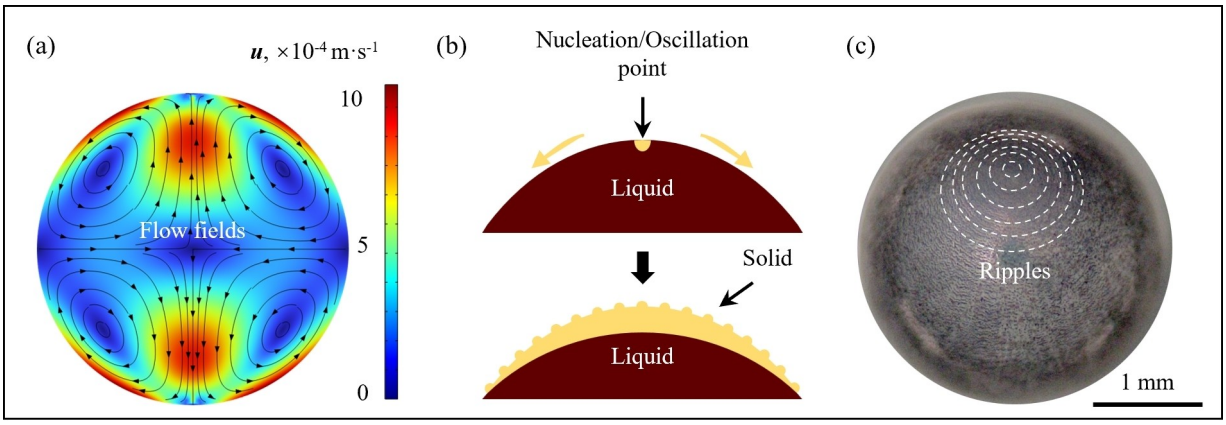

Figure 5. Surface ripple microstructure of $Zr_{64}V_{36}$ alloy droplet solidified in CSS. (a) There is a very slow flow inside the alloy droplet ($u \leq 1.06\times10^{-3}\ m\cdot s^{-1}$) because of the disappearance of natural convection. (b) When there exists a nucleation point, the solid-liquid interface will experience relative vibrations due to the disparities in their properties and diffuses in the form of waves. (c) The are many ripples with a wavelength of 25.8 μm on the surface of alloy droplet solidified in outer space (the average diameter of the sample is about 2.749 mm).

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formed within micron scale at alloy surface. Under microgravity, the natural convection process of liquid is hindered and the droplet surface becomes a free surface. In addition, preferential growth occurs in the direction perpendicular to the surface in the process of crystal growth.[59] Therefore, these spiral microstructures are able to grow and maintain on the surface, which displays a special combination of droplet dynamics and eutectic growth.

The CSS-ESL provides ideal conditions to determine liquid properties, as seen above. However, the liquid structure at atomic scale is much more challenging by experimental methods. The accuracy of interatomic potentials is essential for computing liquid structure. But there is no classical Molecular Dynamics (MD) potential for the Nb–Si system. Here, we use MD based on Deep Neural Networks (DNN) to study liquid structure of $Nb_{82.7}Si_{17.3}$ alloy (Supporting Information, Equation 13–16). It exhibits comparisons between the energies predicted by DNN potential E_{DNN} and energies obtained by ab initio calculations $E_{ab\,initio}$ (Figure 6a). The root mean square errors of energies and forces ($RMSE_e$ and $RMSE_f$) are 3.15×10^{-3} eV·atom^{-1} and 0.29 eV Å^{-1}, respectively, demonstrating the DNN potential predicts appropriate energies and forces. The calculated density at T_E is 5.53 % larger than that obtained in CSS (Figure 6b), which is quite acceptable from the simulation method. Correspondingly, the solute diffusion coefficients of Nb atoms (D_{Nb}) and Si atoms (D_{Si}) are depicted in Figure 6c. It follows the Arrhenius relationship at small undercoolings, but deviates from its at large undercoolings.

The liquid structure of $Nb_{82.7}Si_{17.3}$ alloy was subsequently analyzed by the pair distribution functions (PDF) and Voronoi polyhedron (VP) method.[60–61] The first peak of $PDF_{All\text{-}All}$ increases with decreasing temperature, indicating the ordered degree is enhanced with the increase of undercooling (Figure 6d). The peaks last for seven to eight neighbors at the lowest temperature, suggesting the short-to-medium range order preserves in more than ten angstroms. The $PDF_{Nb\text{-}Si}$ value has visible first peaks among the PDFs. The large peak value of $PDF_{Nb\text{-}Si}$ indicates Nb–Si bonds are favored in the liquid state. The $PDF_{Si\text{-}Si}$ exhibits unique patterns. Its first peak decreases with decreasing temperature. Notably, a small shoulder appears at the second peak of $PDF_{All\text{-}All}$, which is mainly contributed by the second neighbor around Si atoms, not like the first PDF peak. The VPs whose fractions are larger than 2 % are summarized in Figure 6e. The liquid alloy is dominated by Nb-centered VPs with a coordination number (*CN*) of 12–14. Only fractions of <0,2,8,0> and <0,2,8,1> Si-centered VPs are larger than 2 %. The most popular VPs in the liquid alloy are <0,2,8,4>, <0,1,10,2>, <0,1,10,4>, and <0,1,10,3>. Their topologies and surrounding neighbors are depicted in Figure 6f. Some VPs, including <0,3,6,4>, <0,3,6,5>, and <0,3,6,6>, exhibit a trend of increasing firstly and then decreasing with the decreasing temperature.

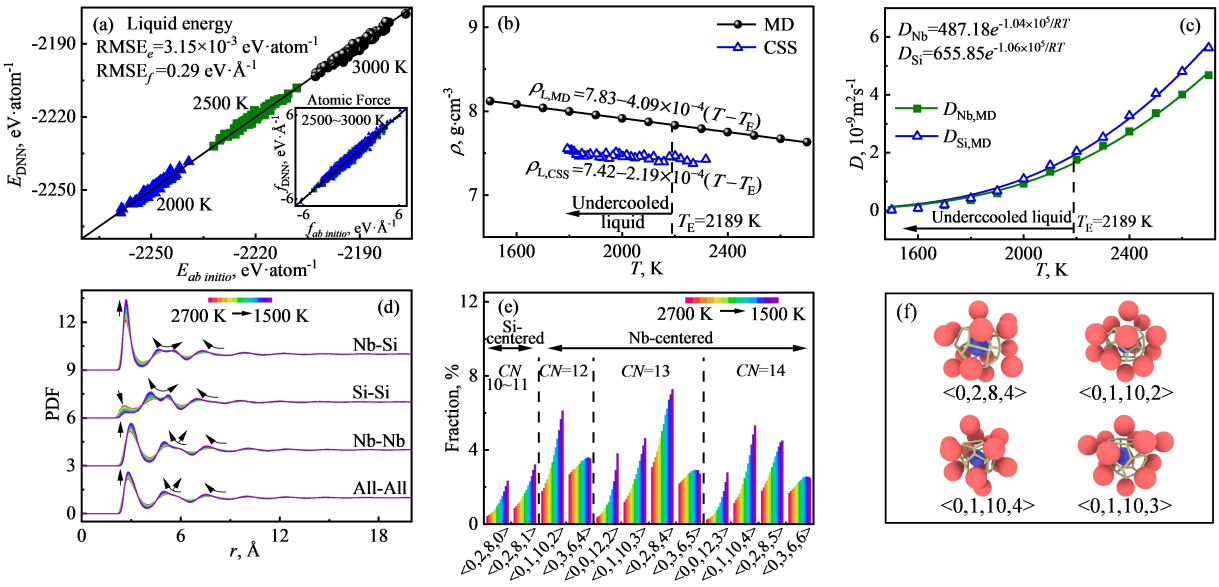

Figure 6. Chemical bond and atomic structure of liquid $Nb_{82.7}Si_{17.3}$ alloy. (a) Energies and forces obtained by DNN potential and ab initio calculations. The root mean square errors of energies ($RMSE_e$) and forces ($RMSE_f$) are 3.15×10^{-3} eV·atom^{-1} and 0.29 eV Å^{-1}, respectively, which demonstrates the DNN potential predicts appropriate energies and forces. (b) The liquid density versus temperature was measured under microgravity aboard the CSS can be described as $\rho_{L,CSS}=7.42–2.19\times10^{-4}(T\text{-}T_E)$. The liquid density versus temperature was calculated by MD can be described as $\rho_{L,MD}=7.83–4.09\times10^{-4}(T\text{-}T_E)$. (c) Solute diffusion coefficients versus temperature. The solute diffusion coefficients of Nb atoms (D_{Nb}) and Si atoms (D_{Si}) were calculated by MD, which both exhibit an exponential relationship with temperature. (d) The liquid structure was subsequently analyzed by the pair distribution functions (PDF) PDFs at different temperatures (2700–1500 K represented using different colored lines). The first peak of $PDF_{All\text{-}All}$, $PDF_{Nb\text{-}Nb}$ and $PDF_{Nb\text{-}Si}$ increases with decreasing temperature, indicating the ordered degree is enhanced. While, the first peak of $PDF_{Si\text{-}Si}$ decreases with decreasing temperature. (e) The VP fractions in liquid state were calculated at different temperatures (2700–1500 K represented using different colored lines). The liquid alloy is dominated by Nb-centered VPs with a coordination number (*CN*) of 12–14. (f) The topologies of most popular VPs in the liquid alloy (<0,2,8,4>, <0,1,10,2>, <0,1,10,4>, and <0,1,10,3>) are depicted.

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These are predominating VPs in face-centered cubic (FCC) crystals,[62–63] suggesting undercooled liquid $Nb_{82.7}Si_{17.3}$ alloy inhibits the formation of FCC-like structures.

Conclusion

The maximum undercooling of 437 K demonstrated that liquid refractory Nb–Si alloy, a very promising candidate material for the fifth generation of aerospace turbine blades, showed a strong undercooling capability to approach even the $0.2T_E$ threshold for homogeneous nucleation. Our findings reveal that the properties, densities and ratio of specific heat to radiative emissivity, were measured with best accuracy, which will be fundamentally important for the future space chemistry manufacture.

Usually for a molten alloy with large viscosity, not for a dilute solution like lake-water, it is assumed that surface ripples on the liquid surface are hard to be kept and the solidified shape could not be changed by the dendritic growth. On the contrary, our experiments indicate that the single point surface nucleation under substantial undercooling and anisotropy volume shrinkage during liquid/solid phase transition resulted in a special olivary shape for solidified $Nb_{82.7}Si_{17.3}$ alloy. A spiral surface microstructure in $Zr_{64}V_{36}$ alloy revealed the special eutectic growth mechanism under Marangoni convection effect. We can conclusively deduce that the novel surface wave patterns prevailing on floating droplets under microgravity condition. These microgravity experiments contribute to a further understanding in high temperature chemistry, space materials chemistry, and solution chemistry.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no competing financial interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: liquid properties · outer space · alloys · computational chemistry

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